MIL-R-60897(MU)
26 June 1968

MILITARY SPECIFICATION
REMOVER, AIRCRAFT CANOPY: M8A1
METAL PARTS ASSEMBLY
(CONVERSION)

1. SCOPE

1.1 This specification covers the overhaul and modification requirements to convert Remover, Aircraft Canopy: M8 metal parts assembly, to Remover, Aircraft Canopy: M8A1 metal parts assembly. This assembly will ultimately be loaded and used in conjunction with other propellant actuated devices in aircrew emergency escape systems.

1.2 Items to be modified under this specification to the M8A1 Remover configuration shall include the T18, T18E1 and M8 Remover.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

QQ-S-634	- Steel, Bar, Carbon, Cold Finished (Standard Quality)
QQ-S-637	- Steel, Bar, Carbon Cold Finished (Standard Quality, Free Machining)
QQ-P-416	- Plating, Cadmium (Electro deposited)

Military

MIL-A-2550	- Ammunition and Special Weapons, General Specification for
MIL-C-5541	- Chemical Films and Chemical Film Materials for Aluminum and Aluminum Alloys

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes

FSC 1377

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Special Tools List

2.1 STANDARDS (Cont'd)

Military

MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-1167 - Ammunition Data Card

DRAWINGS

Munitions Command

C8837890 - Carton, Packing, Ammunition for Remover, Aircraft Canopy, M8 with Cartridge M150
C8837891 - Box, Packing, Ammunition for Remover, Aircraft Canopy, M8 with Cartridge M150
F10522121 - Remover, Aircraft Canopy, M8A1 Metal Parts Assembly (Conversion)
IEL-10522121 - Inspection Equipment List for Remover, Aircraft Canopy, M8A1 Metal Parts Assembly

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bid or request for proposal shall apply.

ASA B46.1 - Surface Texture

(Application for copies should be addressed to the American Standards Association, 10 East 40th Street, New York, N.Y. 10016)

3. REQUIREMENTS

3.1 General.- Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings and referenced specifications.

3.2 Hydrostatic pressure.- The modified outside tube, C10522032 shall withstand a hydrostatic pressure of 2,500 pounds per square inch (psi) for 15 seconds minimum without deformation or mechanical failure (see 6.2).

3.3 Manual function.- Without electrical power and with the clutch disengaged, the pull force required to extend the inside tube assembly shall not exceed 40 pounds.

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3.4 Reversibility load.- The remover shall be capable of withstanding a load of 3060 pounds minimum in tension and 1740 pounds minimum in compression at a stroke of 9.38 inches, reducing linerary to 1490 pounds minimum in tension and 1240 pounds minimum in compression at zero stroke, applied to the remover mounting points. The remover shall not travel due to the applied load.

3.5 Operating performance.- The remover shall be cycled and shall be capable of operating to the loads described by the load curves A, B and C of Figure 1 (see 6.3) and shall comply with the following:

3.5.1 The motor running current drain under load curve B of Figure 1 shall not exceed 13 amperes at 26 to 30 volts D.C. at ambient condition.

3.5.2 The motor running current drain under load curve C shall not exceed 20 amperes at 26 to 30 volts D.C. at ambient condition.

3.5.3 Time required for linear extension of the remover for a normal stroke of 9.38 inches and under load curve B of Figure 1 shall not exceed 12 seconds using the power requirement specified in 3.5.1.

3.5.4 Time required for linear extension of the remover for a normal stroke of 9.38 inches and under load curve C of Figure 1 shall not exceed 15 seconds using the power requirement specified in 3.5.2.

3.5.5 Time requirement for linear retraction of the remover for a normal stroke of 9.38 inches under load Curve A of Figure 1 shall not exceed 15 seconds at 26 to 30 volts D.C. with a running current drain not to exceed 15 amperes at ambient condition.

3.5.6 When the load aids the operation of the remover for the full stroke of 9.38 inches, the operating time shall be 5 seconds minimum to 10 seconds maximum and the running current shall not exceed 10 amperes at 26 to 30 volts D.C.

3.5.7 There shall be no structural failure of any component under the inertia loads imposed by the sudden interruption of electrical power.

3.5.8 At the end of the duty cycle for normal load curve B, Figure 1, the remover shall pull against the stop with a force, known as cinch-down load, of 1000 to 1450 pounds. Under this cinch-down load requirement, the load limit switch shall de-energize the remover electric motor.

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3.5.9 Time required to engage clutch shall not exceed 500 milliseconds (See 6.4).

3.5.10 After the remover has cinched-down, the clutch shall be disengaged. The time required to disengage the clutch shall not exceed 500 milliseconds (See 6.5).

3.6 Breakaway torque.- The assembled unit shall withstand the following breakaway torque without movement or deformation.

TABLE I

Components	Breakaway Torque (inch pounds)
Tube, Outside to Adapter Housing	300 Min.
Plug to Tube, Inside	400 Min.
End Rod to Plug	400 Min.

3.7 Metal parts assembly.- Parts shall be combined to produce an assembly which conforms to the requirements of the metal parts assembly drawing and the following:

3.7.1 The conversion metal parts assembly shall contain the following newly designed components and sub-assemblies:

Nomenclature	Drawing Number	Number Required
Housing	F10522018	1
Base	D10522017	1
Tube, Inside	D10522036	1
Saddle	D10522094	1
Bracket, Switch, Up Limit	D10522105	1
Cover, Switch, Up Limit	D10522122	1
Cable Assembly	D10522123	1
Retainer, Ball	D10522136	1
Switch, Up Limit (Normally Closed)	D10522132	1
Tang, Retainer	D10543078	1
Retainer	C10522019	1
Lever, Torque Limit	C10522034	1
Pawl	C10522051	1
Retainer, Piston	C10522058	2
Gasket	C10522071	1

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3.7.1 (Cont'd)

Nomenclature	Drawing Number	Number Required
Holder, Wire	C10522075	1
Spring	C10522095	1
Rod, Switch Actuating	C10522098	1
Spring, Rod Actuator	C10522099	1
Nut	C10522102	1
Nut, Adjusting	C10522103	1
Cap, Bearing Holder	C10522107	1
Holder, Bearing	C10522108	1
Bearing, Seal Motor	C10522126	1
Screw, Machine	B10522002	1
Shaft, Torque Limit	B10522004	1
Cover	B10522027	1
Spacer	B10522093	1
Seal	B10522128	1
Pin	A10522069	1
Washer	B10522133	2
Ball (Steel)	MS19060-3	1
Clamp	MS35842-5	1
Grommet	MS35489-4	1
Grommet	MS35489-67	2
Grommet	MS35489-93	1
Nut, Self Locking	AN345-1	1
Pin, Spring	MS20364-624C	1
Seal, "O" Ring	MS9048-003	1
Seal, "O" Ring	MS9021-004	1
Seal, "O" Ring	MS9021-006	1
Seal, "O" Ring	MS9021-031	1
Screw, Machine Cap	MS28775-008	1
Screw, Machine	MS16997-9	2
Screw, Machine	MS51957-7	2
Screw, Machine	MS51957-15	10
Screw, Machine	MS51957-16	2
Screw, Machine	MS51957-30	5
Screw, Machine	MS51957-32	4
Screw, Machine	MS51959-9	2
Screw, Set	MS51974-1	1
Screw, Set	MS51963-1	1
Splice, Electric	MS25182-2	5
Washer, Flat	MS15795-4	1
Washer, Flat	MS27183-1	1
Washer, Flat	MS27183-4	2
Washer, Lock	MS35338-39	1
Washer, Lock	MS35338-40	2

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3.7.2 The conversion metal parts assembly shall contain the following components which shall be modified to the applicable modification drawing.

Nomenclature	Original Drawing	Modification Drawing No.	New Replacement Drawing No. (See 3.7.2.1)
Mount, Motor	FC9475	D10522001	F10522003
Gear Box	FF9838	D10522006	F10522008
Tube, Outside	FC9476	C10522032	D10522060
Drum, Clutch	FB51665	C10522042	C10522119
Cup	FC9480	C10522043	C10522118
Race, Outer	FB51792	B10522077	C10522026
Lever	FB51679	B10543079	C10522012

3.7.2.1 New production parts shall be in accordance with the applicable replacement drawing.

3.7.2.2 The Gear Box and Motor Mount Assembly which were manufactured to the original drawing FF9838 shall be retained as a matched set.

3.7.3 The remover metal parts assembly shall contain the following new parts. These parts which are to be replaced are considered 100 percent throw-away items.

Name of Part	Original Drawing	New Production Drawing
Sleeve (Bulk Material)	FA29985 (P/N FA29985-A)	FA29985 (P/N FA29985-A
Actuator Switch (2 req'd)	FA29928	C10522135
Detent Assembly	FB51687	B10522055
"O" Ring (2 req'd)	FA29952	MS28775-008
Pin, Shear	FB51632	A10522065
Plug, Shipping	A8593299	A8593299
Screw, Shear (4 req'd)	FC9752	B10522062
Screw	FB51683	B10522061
Spring, Detent Lever	FA29923	C10522013
Spring	FA29948	B10522090
Spring, Piston (2 req'd)	FB51631	B10522068
Spring, Clutch	FB51684	B10522110
Packing "O" Ring	AN6227-7	
Packing "O" Ring	MS28775-013	
Packing "O" Ring	MS28775-332	
Packing "O" Ring	MS29513-127	
Packing "O" Ring	MS29513-134	
Packing "O" Ring	MS29513-142	
Packing "O" Ring	MS29513-145	
Clamp, Cushion Support (2 req'd)	MS21919-H4	

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3.7.4 In the conversion of the M8 Remover to the M8A1 Remover, the following components shall be deleted and discarded except as noted:

Nomenclature	Drawing Number
Housing	FD20732
Spring, Helical, Ext.	FC9743
Lever, Torque, Limit	FC9744
Pin, Shear	FB51630
Cam	FB51649
Screw	FB51651
Worm	FB51653
Gear	FB51654
Gear	FB51655
Mount	FB51658
Cover	FB51662
Gear	FB51664
Ring	FB51669
Label	FB51770
Bracket, Shipping	FB51675
Trigger	FB52186
Screw	FB51769
Eye	FB52221
Pin	FA29918
Tube, Inside	FC9478
Mount, Lever Assembly *	FC9484
Washer	FB51680
Cover	FB51661
Gear Cluster	FB51663
Gear Cluster Assembly	FB51668
Switch, Up Limit	FB51678
Bushing	FA29949
Shaft, Worm	FA29935
Collar	FA29946
Shaft, Cam	FA29941
Clamp	FA29947
Lever, Mount	FA29925
Bracket	FA29910
Actuator	FA29924
Sleeve	FA29985B
Sleeve	FA29985C
Sleeve	FA29985D
Washer	FA29986
Pad	FA29929
Slide	FA29951

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3.7.4 (Cont'd)

Nomenclature	Drawing Number
Washer	MS15795-308
Washer, Flat	MS15795-304
Washer, Lock **	MS35377-77
Screw	MS35233-28
Clamp, Cushion, Support ***	MS21919-H4
Clamp, Cushion, Support	MS21919-H5
Screw	MS35237-21
Pin, Spring	MS9048-036
Pin, Spring	MS9048-034
Pin, Spring	MS9048-104
Pin	AN393-9
Pin	AN380-2-2
Ring	NAS51-18
Ring	NAS50-112

* Retain in part
** Retain 4
*** Retain 2

3.7.5 Bolts, nuts, screw inserts or any other thread components which show evidence of cross-threading, stripping or excessive wear shall be replaced.

3.7.6 Bearing, ball bearing screw assemblies, and ball bearings which show evidence of roughness, binding or damage shall be replaced.

3.7.7 Switches which contain a loose plunger, damaged wiring, gummy or defective plunger seals shall be replaced.

3.7.8 Electric motor shall be replaced if it is unrepairable.

3.7.9 Any gear with excessive backlash, worn or broken teeth or improper mating shall be replaced with a new gear.

3.7.10 Gear box and ballistic assembly.- Any parts in the Gear Box and Ballistic Assembly which show evidence of distortion, excessive wear, looseness or any other damage shall either be repaired or replaced with a new part.

3.7.11 Gear box lubrication.- The grease in the gear box cavity shall be removed and replaced with fresh grease conforming to the requirements of the applicable grease specification.

3.7.12 Spring pins shall be replaced if removed from an assembly or if found defective.

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3.8 Protective finish.- Protective finish shall be in accordance with the applicable specifications and drawings and shall apply to all surfaces of the component regardless of configuration. All components undergoing modification or remanufacture shall have the existing protective finish stripped, and upon completion of rework, the specified protective finish re-applied (see 6.6 through 6.6.5).

3.9 Workmanship.- The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.9.1 Metal defects.- Components shall be free from cracks, splits, or other metal defects.

3.9.2 Burr.- No part shall have a burr which might interfere with the assembly or function of the item, or which might be injurious to personnel handling the item.

3.9.3 Foreign matter.- Chips, dirt, grease (except where required), rust, corrosion or other foreign matter shall be removed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define the Quality Assurance terms used.

4.1.2 Inspection.- Inspection shall be in accordance with MIL-A-2550.

4.2 Inspection provisions.

4.2.1 Lot.

4.2.1.1 Submission of product.- The product shall be submitted in accordance with MIL-STD-105.

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4.2.1.2 Size of lot.- For the purpose of identification and shipment, the shipping lots of metal parts assemblies shall consist of 1200 removers, maximum, exclusive of test samples, provided the homogeneity of remanufacturing process has been retained throughout the inspection lot. A smaller lot may be shipped upon approval of the contracting officer.

4.2.2 Examination.- The roughness comparison specimens prescribed by ASA B46.1 shall be used as a basis of comparison for surface roughness determination. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

4.2.2.1 Reclaimed parts.- Examination of reusable parts, which require no reprocessing shall be limited to inspection of defects that require visual type inspection. One hundred percent examination shall be performed for all such defects. All nonconforming parts shall be rejected.

4.2.2.2 Reprocessed parts.- One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plan and associated criteria in Tables II and III. For an individual defect (single characteristic) the criteria in Table II shall be used. For total defects within a major or minor group, the criteria in Table III shall be used. All nonconforming material shall be rejected.

TABLE II

Lot Size	Sample Size	Major		Minor	
		Accept	Reject	Accept	Reject
1 - 300	100% Examination				
301 - 1300	110	1	2	2	3
1301 - 3200	150	2	3	3	4
3201 - 8000	225	3	4	4	5
8001 - 22000	300	4	5	5	6

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4.2.2.2 (Cont'd)

TABLE III

Total Allowable Defects Per Sample Size

Sample Sizes for Major Defects	110	150	225	300
Number of Defects in Class				
2 - 3	1	2	3	4
4 - 7	2	3	4	5
8 - 19	3	4	5	7
20 - 43	4	5	8	10
Over - 43	6	8	11	14

Sample Sizes for Minor Defects	110	150	225	300
Number of Defects in Class				
2 - 3	2	3	4	5
4 - 7	3	4	5	7
8 - 16	4	5	8	10
17 - 48	6	8	11	14
Over - 48	8	11	17	20

4.2.2.3 Head, Firing, Drawing D10522045 covering a detail of Drawing F10522124

Defect	Method of Inspection
CRITICAL	
1. Metal defective	Visual
MAJOR	
101. Pitch diameter of small inside thread, max.	Gage
102. Minor diameter of small inside thread, max.	Gage
103. Pitch diameter of large inside thread, max.	Gage
104. Minor diameter of large inside thread, max.	Gage
105. Diameter of long bore	Gage
106. Protective finish inadequate or defective	Visual
107. Burr	Visual
108. Foreign matter	Visual

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4.2.2.3 (Cont'd)

Defect	Method of Inspection
MINOR	
None defined	

4.2.2.4 Tube, Outside Assembly, Drawing C10522032 covering a detail of Drawing F10522124

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
101. Pitch diameter of external thread, min.	Gage
102. Major diameter of external thread, min.	Gage
103. Diameter of hole	Gage
104. Diameter of counterbore	Gage
105. Depth of counterbore	Gage
106. Assembly modification incomplete	Gage
107. Metal defective	Visual
108. Protective finish inadequate or defective	Visual
109. Burr	Visual
110. Foreign matter	Visual
MINOR	
None defined	

4.2.2.5 Ring, Locking, Drawing B10522063 covering a detail of Drawing F10522124

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
101. Pitch diameter of inside thread, max.	Gage
102. Minor diameter of inside thread, max.	Gage
103. Metal defective	Visual
104. Protective finish inadequate	Visual
105. Burr	Visual
106. Foreign matter	Visual
MINOR	
None defined	

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4.2.2.6 Cover, Drawing B10522140 covering a detail of Drawing F10522124

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
101. Metal defective	Visual
102. Protective finish inadequate or defective	Visual
103. Burr	Visual
104. Foreign matter	Visual
MINOR	
None defined	

4.2.2.7 End, Rod, Drawing C10522049 covering a detail of Drawing F10522124

CRITICAL	
None defined	
MAJOR	
101. Pitch diameter of outside thread, min.	Gage
102. Minor diameter of outside thread, min.	Gage
103. Diameter of bore, max.	Gage
104. Metal defective	Visual
105. Protective finish inadequate or defective	Visual
106. Burr	Visual
107. Foreign matter	Visual
MINOR	
None defined	

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4.2.2.8 Plate, Cover, Drawing B10522024 covering a detail of Drawing F10522124

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
101. Metal defective	Visual
102. Protective finish inadequate or defective	Visual
103. Burr	Visual
104. Foreign matter	Visual
MINOR	
None defined	

4.2.2.9 Retainer, Bearing, Drawing B10522050 covering a detail of Drawing F10522124

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
101. Metal defective	Visual
102. Protective finish inadequate or defective	Visual
103. Burr	Visual
104. Foreign matter	Visual
MINOR	
None defined	

4.2.2.10 Housing, Left Bearing, Drawing B10522088 covering a detail of Drawing F10522124

CRITICAL

None defined

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4.2.2.10 (Cont'd)

Defect	Method of Inspection
MAJOR	
101. Diameter of small diameter, min.	Gage
102. Diameter of counterbore, max.	Gage
103. Metal defective	Visual
104. Protective finish inadequate or defective	Visual
105. Burr	Visual
106. Foreign matter	Visual
MINOR	
None defined	

4.2.2.11 Plug, Inside Tube, Drawing C10522041 covering a detail of Drawing F10522124

CRITICAL	
None defined	
MAJOR	
101. Pitch diameter of inside thread, max.	Gage
102. Minor diameter of inside thread, max.	Gage
103. Pitch diameter of outside thread, min.	Gage
104. Major diameter of outside thread, min.	Gage
105. Diameter of spring pin hole, max.	Gage
106. Spring pin holes, unplugged	Visual
107. Metal defective	Visual
108. Protective finish inadequate or defective	Visual
109. Burr	Visual
110. Foreign matter	Visual
MINOR	
None defined	

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4.2.2.12 Cover, Drawing B10522016 covering a detail of Drawing F10522124

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
101. Metal defective	Visual
102. Protective finish inadequate or defective	Visual
103. Burr	Visual
104. Foreign matter	Visual
MINOR	
None defined	

4.2.2.13 Ring, Shear, Drawing C10522048 covering a detail of Drawing F10522124

CRITICAL	
None defined	
MAJOR	
101. Outside diameter, min.	Gage
102. Metal defective	Visual
103. Protective finish inadequate or defective	Visual
104. Burr	Visual
105. Foreign matter	Visual
MINOR	
None defined	

4.2.2.14 Housing, Main Bearing, Drawing C10522039 covering a detail of Drawing F10522124

CRITICAL	
None defined	

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4.2.2.14 (Cont'd)

Defect	Method of Inspection
MAJOR	
101. Diameter of large bore, max.	Gage
102. Diameter of small bore, max.	Gage
103. Diameter of large outside shoulder, min.	Gage
104. Metal defective	Visual
105. Protective finish inadequate or defective	Visual
106. Burr	Visual
107. Foreign matter	Visual
MINOR	
None defined	

4.2.2.15 Adapter Housing, Drawing D10543246 covering a detail of Drawing F10522124

CRITICAL	
None defined	
MAJOR	
101. Pitch diameter of inside thread, max.	Gage
102. Minor diameter of inside thread, max.	Gage
103. Pitch diameter of outside thread, min.	Gage
104. Major diameter of outside thread, min.	Gage
105. Metal defective	Visual
106. Protective finish inadequate or defective	Visual
107. Metal defective	Visual
108. Burr	Visual
109. Foreign matter	Visual
110. Pellet missing, loose or improperly installed	Visual
MINOR	
None defined	

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Glossary, shall be prepared in accordance with MIL-M-38784(MU).

4.2.2.16 Ball Bearing, Screw Assembly, Drawing 10522056 covering a detail of Drawing F10522124

	Defect	Method of Inspection
CRITICAL		
	None defined	
MAJOR		
101.	Any part missing	Visual
102.	Modification missing or incorrect	Visual
103.	Protective finish inadequate or defective	Visual
104.	Metal defective	Visual
105.	Burr	Visual
106.	Foreign matter	Visual
MINOR		
	None defined	

4.2.2.17 Mount, Motor, Drawing D10522001 covering a detail of Drawing F10522124

	Defect	Method of Inspection
CRITICAL		
	None defined	
MAJOR		
101.	Pitch diameter of tapped holes, max. (4 places)	Gage
102.	Minor diameter of tapped holes, max. (4 places)	Gage
103.	Location of tapped holes incorrect (4 places)	Gage
104.	Pitch diameter of tapped holes, max. (2 places)	Gage
105.	Minor diameter of tapped holes, max. (2 places)	Gage
106.	Location of tapped holes incorrect (2 places)	Gage
107.	Depth of tapped holes, incorrect	Gage
108.	Diameter of large hole incorrect (modification)	Gage
109.	Location of large hole incorrect	Gage

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4.2.2.17 (Cont'd)

Defect	Method of Inspection
MAJOR (Cont'd)	
110. Plug not flush with surface (modification)	Gage
111. Metal defective	Visual
112. Protective finish inadequate or defective	Visual
113. Burr	Visual
114. Foreign matter	Visual
MINOR	
None defined	

4.2.2.18 Drum, Clutch, Drawing C10522042 covering a detail of Drawing F10522124

CRITICAL	
None defined	
MAJOR	
101. Diameter of hole incorrect	Gage
102. Location of hole incorrect	Gage
103. Diameter of counterbore incorrect	Gage
104. Depth of counterbore incorrect	Gage
105. Diameter of hole in hub incorrect	Gage
106. Outside diameter incorrect	Gage
107. Metal defective	Visual
108. Protective finish inadequate	Visual
109. Burr	Visual
110. Foreign matter	Visual
MINOR	
None defined	

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4.2.2.19 Cup, Drawing C10522043 covering a detail of Drawing F10522124

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
101. Width of cutout incorrect	Gage
102. Depth of cutout incorrect	Gage
103. Length of cutout incorrect	Gage
104. Radius of cutout incorrect	Gage
105. Metal defective	Visual
106. Protective finish inadequate	Visual
107. Burr	Visual
108. Foreign matter	Visual
MINOR	
None defined	

4.2.2.20 Race, Outer Drawing B10522077 covering a detail of Drawing F10522124

CRITICAL	
None defined	
MAJOR	
101. Width incorrect	Gage
102. Diameter of countersunk hole incorrect	Gage
103. Location of countersunk hole	Gage
104. Metal defective	Visual
105. Protective finish inadequate	Visual
106. Burr	Visual
107. Foreign matter	Visual
MINOR	
None defined	

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4.2.2.21 Lever, Drawing B10543079 covering a detail of Drawing F10522124

Defect		Method of Inspection
CRITICAL		
	None defined	
MAJOR		
101.	Width incorrect	Gage
102.	Surface finish incorrect	Gage
103.	Metal defective	Visual
104.	Protective finish inadequate	Visual
105.	Burr	Visual
106.	Foreign matter	Visual
MINOR		
	None defined	

4.2.2.22 Box, Gear, Drawing D10522006 covering a detail of Drawing F10522124

Defect		Method of Inspection
CRITICAL		
	None defined	
MAJOR		
101.	Pitch diameter of tapped holes, max.	Gage
102.	Minor diameter of tapped holes, max.	Gage
103.	Location of tapped holes incorrect	Gage
104.	Diameter of large bore incorrect	Gage
105.	Depth of large bore from face incorrect	Gage
106.	Metal defective	Visual
107.	Protective finish inadequate or defective	Visual
108.	Burr	Visual
109.	Foreign matter	Visual
MINOR		
	None defined	

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4.2.2.23 Metal Parts Assembly, Drawing F10522124

Defect	Method of Inspection
CRITICAL	
1. Length of firing pin striker	Gage
2. Improper or insecure assembly	Manual and Visual
3. Loose or improper wiring	Manual and Visual
4. Improper soldering	Visual
5. Gas port hole obstructed	Visual
6. Any part missing	Visual
7. Marking missing, incorrect or illegible	Visual
8. Burr	Visual
9. Foreign matter	Visual
MAJOR	
None defined	
MINOR	
None defined	

4.2.2.24 Packing in Inner Box, Drawing C8837890

CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Contents loose	Manual
202. Top of box loose	Manual
203. Strapping missing, loose or improperly assembled	Manual and Visual
204. Supports or fillers missing from box	Visual
205. Box damaged	Visual
206. Seal missing or improperly positioned	Visual
207. Motor wires loose	Visual
208. Marking missing, incorrect or illegible	Visual

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4.2.2.25 Packing in Outer Box, Drawing C8837891

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Contents loose	Manual
202. Liner, fillers, cushings or spacers missing	Visual
203. Strapping missing, loose or improperly assembled	Visual
204. Hardware missing, loose or improperly engaged	Visual
205. Box damaged	Visual
206. Contents of box exposed	Visual
207. Car seal missing or improperly sealed	Visual
208. Marking missing, incorrect or illegible	Visual

4.2.3 Inspection.- The following tests shall be performed in accordance with 4.3.

4.2.3.1 Hydrostatic pressure.- Each outside tube shall be subjected to the hydrostatic pressure test. Failure of any item to comply with the requirements of 3.2 shall be cause for rejection of the item.

4.2.3.2 Protective finish, replated components.- The number of specimens and acceptance criteria for protective finish tests shall be in accordance with the applicable Military Standards listed on the drawing.

4.2.3.3 Breakaway torque.- A breakaway torque shall be applied to each assembly threaded joint as listed in Table I. Failure of any assembly to comply with the requirements of Table I shall be cause for rejection of the assembly.

4.2.3.4 Manual function.- Each remover shall be tested for the pull force required to extend the inside tube. Failure of any remover to comply with the requirements of 3.3 shall be cause for rejection of the remover.

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4.2.3.5 Reversibility load.- Each remover shall be tested for reversibility load. Failure of any remover to comply with the requirement of 3.4 shall be cause for rejection of the remover.

4.2.3.6 Operating performance.- Each remover shall be cycled electrically 5 times each to the load curves of Figure 1. Failure of the remover to comply with the applicable requirements of 3.5 shall be cause for rejection.

4.2.4 Inspection equipment.- The examination and tests shall be made using equipment listed on IEL-10522124.

4.3 Test methods and procedures.

4.3.1 Hydrostatic pressure.- Prior to the application of protective finish, the outside tube shall be mounted in the test device and subjected to an internal hydrostatic pressure to determine compliance with the requirements of 3.2. The test device used shall internally seal the tube within $1\frac{1}{2}$ inches of each end without applying an axial load other than that applied by the liquid pressure.

4.3.2 Protective finish, replated components.- Test procedures for protective finish shall be in accordance with the applicable Military Standard referenced on the drawing.

4.3.3 Breakaway torque.- A torque wrench set at the torque requirement specified in Table I for the respective assembly or sub-assembly shall be used to determine the adequacy of the joint. The test shall be accomplished by securing the assembly or subassembly in a fixture.

4.3.4 Manual function.- The inside tube shall be extended manually with the clutch disengaged and without electric power as specified in 3.3.

4.3.5 Reversibility load.- The test shall be performed at zero inches, midpoint of stroke and at full stroke with the load in tension and compression as specified in 3.4.

4.3.6 Operating performance.- The remover shall be secured in the test fixture between remover mounting points and tested in accordance with 4.2.3.6. The indicated values obtained shall be recorded and used to determine compliance with 3.5.

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5. PREPARATION FOR DELIVERY

5.1 Packing and marking (Level A).- Packing and marking shall be in accordance with Drawings C8837890 and C8837891. Marking shall be changed to specify Remover, Aircraft Canopy, M8A1.

5.2 Data cards.- Data card information shall be as required in specification MIL-STD-1167.

6. NOTES

6.1 Ordering data.- Procurement documents should specify the title, number and date of this specification.

6.2 A mechanical failure is defined as any deformation or breakage of a part, the occurrence of which is other than a design function.

6.3 A complete electrical operating cycle consists of clutch engagement, extension, retraction, cinch-down, and clutch disengagement.

6.4 Clutch engagement occurs with the closing of the up-switch on the control box. The failure to engage the clutch is detected with no cinch-down.

6.5 Clutch disengage time is taken from energizing the disengage solenoid and releasing the cinch-down load to current shut off at disengage solenoid.

6.6 Remanufactured exterior components are listed in Table IV.

TABLE IV

Name of Part	Drawing Number
Ballistic Assembly	
Adapter, Housing	D10543246
Head, Firing	D10522045
Housing, Main Bearing	C10522039
Plug, Inside Tube	C10522041
Ring, Locking	B10522063
Ring, Shear	C10522048
Rod, End	C10522049
Gear Box Assembly	
Cover	B10522140
Cover	B10522016
Cover, Plate	B10522024
Housing, Left Bearing	B10522088
Retainer, Bearing	B10522050

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6.6.1 The outside surfaces of the remanufactured engage and disengage solenoids shall be cleaned with steel wool and a coating of acrylon shall be sprayed on the outside diameter.

6.6.2 Bare spots on the electric motor shall be touched-up with black paint.

6.6.3 Exposure of base metal on all other internal aluminum components shall be treated with a chemical film, Grade A or B, Class 1, Specification MIL-C-5541.

6.6.4 Nylon pellets.- Components which contain nylon pellets and are to be replated shall have the pellets removed and new pellets installed after plating in accordance with the applicable drawing.

6.6.5 Locking inserts.- Components which contain locking inserts and are to be replated shall have the locking inserts removed and new inserts installed after plating in accordance with the applicable drawing.

6.7 Special instructions covering modification, repair and assembly of remanufactured components are as follows:

6.7.1 Plug - (C10522041).- Plugs which contain elongated or damaged spring pin hole on plug boss may be reclaimed by locating an additional hole approximately 45 degrees from the existing hole. Number of holes in plug shall be limited to three (3). The old holes shall be plugged with plastic steel compound (DEVCON).

6.7.2 Ball bearing screw assembly.- (D10522056) Ball bearing assemblies, which contain the stake screw, stop, shall be modified to incorporate the two (2) each Pin, Spring MS9048-073 as required by Drawing D10522056. If new stop screws are required, the part shall comply with dimensions noted on Drawing D10522056. Material and protective finish requirements for this part shall comply with the following:

6.7.2.1 Material.- Steel, Carbon (1000-1030) Normalized Condition, QQ-S-634.

6.7.2.2 Alternative material.- Steel, Carbon (1010 to 1030) Normalized Condition, QQ-S-637.

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6.7.2.3 Protective finish.- Cadmium Plating, Class 2, Type II, with supplementary chromate treatment, QQ-P-416.

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1377-A266

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Verification Evaluation

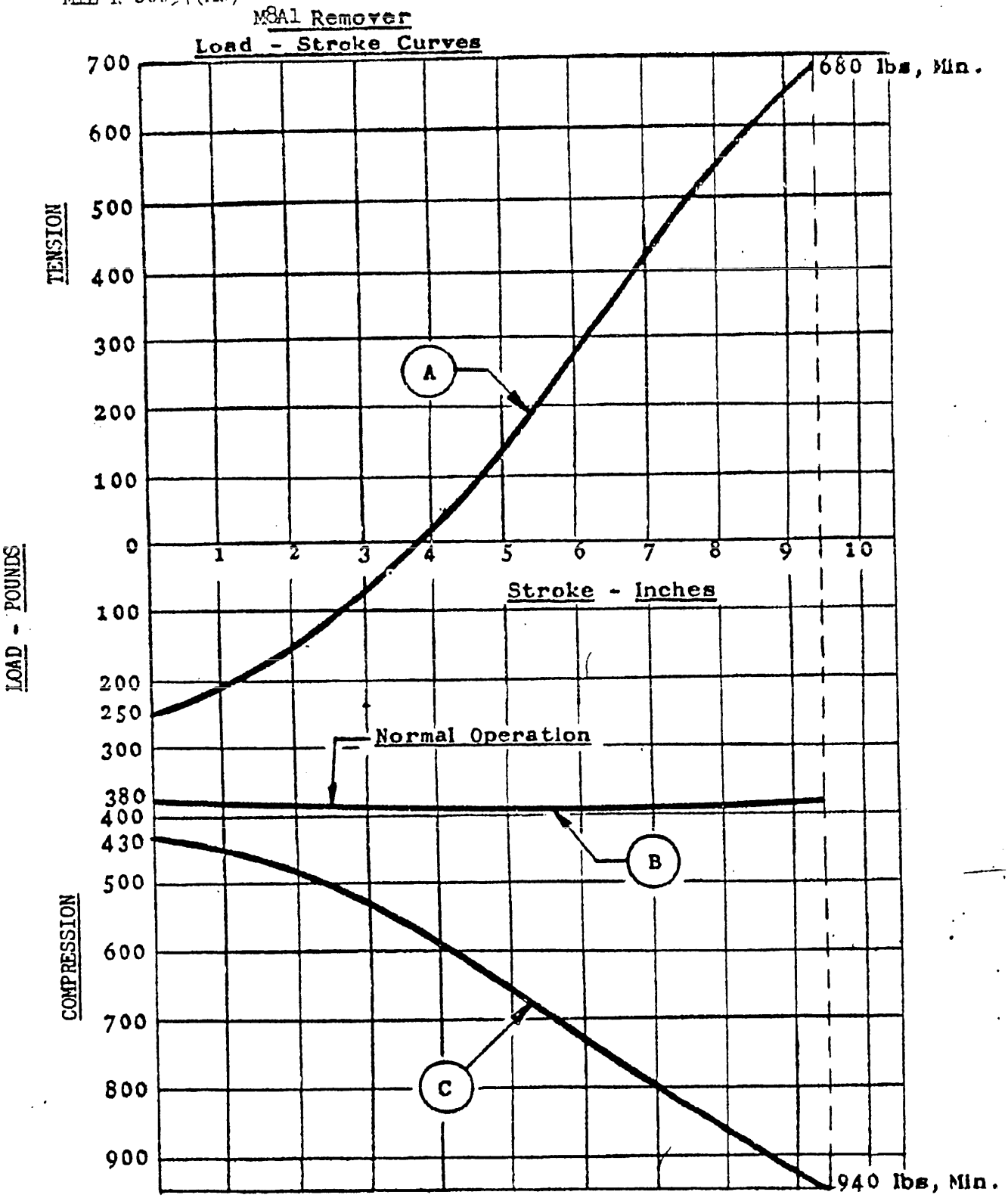


FIGURE 1

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

[ ] YES [ ] NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 68 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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